

IRexp experimental infrared band lists from the PMC Open Access Subset and Chemotion

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Abstract

IRexp is a redistributable collection of **experimental infrared band lists** (cm⁻¹ peak positions) mined from open chemistry literature, optionally with ¹H/¹³C shifts and resolved structures. The release holds **121,233** records (119,345 PMC OA; 1,888 Chemotion/RADAR4Chem), with **43,060** structure-linked and **33,201** full IR + ¹H + ¹³C + structure quadruples. IRexp stores **numeric band lists**, not absorbance traces. Records carry `source_doi` and stamped `license / license_pool`. Licensing is **mixed**: Europe PMC join yields **87,617** commercial (CC-BY/CC0), **20,938** non-commercial (NC*), **10,781** empty/unknown (excluded from commercial Zenodo), and **1,897** ShareAlike (Chemotion + rare PMC SA). Reuse: multimodal training and complementary elucidation benchmarks citing this Descriptor. Data: Hugging Face ilkhamfy/IRexp; Zenodo DOI pending archival mint. Code is MIT-licensed.

1 Background & Summary

Infrared (IR) spectroscopy is routine in organic characterisation, yet **open, redistributable** collections of *experimental* IR data remain sparse relative to modern machine-learning needs. Digitised absorbance libraries such as the NIST Chemistry WebBook [National Institute of Standards and Technology, 2026] and AIST SDBS [National Institute of Advanced Industrial Science and Technology, 2026] are valuable but either modest in size or **view-only** (no bulk redistribution). Computational IR–NMR resources published in *Scientific Data* expand multimodal coverage with simulated spectra [Zipoli et al., 2025], while large literature mines for NMR peak lists (notably NMRexp [Wang et al., 2025]) demonstrate that peer-reviewed experimental spectral *lists* with DOI traceability are in scope for this journal. Concurrent literature corpora that include IR among other modalities (for example NMRSpec/NMRTrans [Yang et al., 2026]) further motivate an **IR-focused, redistributable** band-list resource rather than another closed spectrum archive.

IRexp fills a specific wedge. Experimental sections of chemistry papers conventionally report per-compound **IR band lists** (wavenumbers in cm⁻¹) together with ¹H/¹³C NMR shift lists. That textual convention is the object language models and many elucidation pipelines consume, and it is a **different object** from a digitised spectrum. IRexp therefore:

1. Harvests open-access full text from the **PMC Open Access Subset** bulk S3 mirror and peak lists from the **Chemotion** FT-IR deposit.
2. Extracts deterministic numeric IR band lists (and co-reported NMR strings where present).
3. Resolves compound names to canonical structures with OPSIN [Lowe et al., 2011], RD-Kit [Landrum, 2026], and SELFIES [Krenn et al., 2020] where possible.
4. Releases **extracted numbers only** — no PDFs, figures, or article full text — with source accessions for attribution.

Among openly redistributable *text-derived IR band lists*, IRexp is large by record count (121,233). It does not claim to replace SDBS or NIST absorbance libraries; SDBS alone holds more structure-linked *spectra* than IRexp holds structure-linked band lists. The intended reuse is multimodal pretraining, supervised IR→structure modelling on `irexp_resolved`, and as the literature substrate for complementary elucidation benchmarks described elsewhere (forthcoming ICLR manuscript on IRSpectra-Bench; cite that work for protocol and model results — **not** reproduced here) [Yabbarov et al., 2026].

What this Data Descriptor does not contain. No hypothesis tests, no large-language-model accuracy tables, and no stage-decomposition of elucidation performance. Those belong in the complementary research paper.

2 Methods

2.1 Source corpora

PMC Open Access Subset. Primary harvest uses the NCBI PMC OA bulk distribution on Amazon S3 (`s3://pmc-oa-opendata`, HTTPS endpoint `https://pmc-oa-opendata.s3.amazonaws.com`) [National Library of Medicine, 2026]. **Harvest window (recoverable from git snapshots):** the bulk S3 IR crawl and `seen_papers / ir_harvest_snapshot` artefacts were produced on **2026-06-07 (UTC)** (`scripts/s3_ir_harvest.py`; incremental auto-snapshots that day through ~134,893 raw IR rows). Chemotion was ingested the same day. A harvest snapshot records **188,016** distinct PMC identifiers scanned (`data/irexp/seen_papers.txt.gz`). The released corpus retains records from **15,416** unique PMC:* accessions. Extraction operates on open-access **plain text** objects at flat S3 keys `PMC{id}.*{v}/PMC{id}.*{v}.txt`. Europe PMC full-text XML is used for some validation re-fetches. The **released** IRexp DOIs are exclusively PMC:* accessions or the Chemotion deposit DOI; no paywalled publisher DOI appears in the 121,233-record file.

Discovery vs `oa_comm` / `oa_noncomm` packages. Identifier discovery used NCBI E-utilities `esearch` over PMC with an IR-characterisation query **and open access[filter]**, then fetched matching IDs from the flat S3 text layout. The harvest therefore did **not** pre-filter by walking the PMC OA Subset’s `oa_comm` vs `oa_noncomm` vs other package directories. PMC OA is still **not** a single licence [National Library of Medicine, 2026]. Licence truth for redistribution is applied **post-hoc**: every unique PMCID in IRexp (15,416) was joined to Europe PMC `license` metadata (`scripts/join_pmc_licences.py`); each record carries `license / license_pool`. Commercial-use (CC-BY/CC0) rows form the Zenodo / Sci Data primary pool; NC* are held aside; empty/unknown are **excluded** from the commercial deposit — aligning redistributable intent with the OA commercial/non-commercial distinction via article-level licences rather than S3 package paths (`data/NOTICE`; `docs/scientific_data/LICENCE_REMEDIATION.md`).

Chemotion / RADAR4Chem. **1,888** records come from the Chemotion Repository FT-IR collection deposited at RADAR4Chem (DOI 10.22000/0GoEQGlsZGEIrgst) [Jung et al., 2024], licensed **CC-BY-SA-4.0**. Ingest (`scripts/chemotion_to_irexp.py`, 2026-06-07): download the MD5-verified deposit; for each of **2,116** ATR-IR analyses, flatten the Quill-delta `content` field to plain text; parse an **author-curated** IR band list with the **same** regex extractor and quality gates as the PMC path; resolve the deposit’s canonical SMILES with RDKit → InChIKey + SELFIES; keep the richest band list per InChIKey; drop the **8** molecules already present in the PMC pool → **+1,888** new structure-resolved rows. These are **not** algorithmic peak-picks from absorbance curves; they are author-entered experimental band lists from the

ELN deposit, denser than typical paper prose (median **39** bands vs **9** for PMC). Rows carry `license=CC-BY-SA-4.0` and `source_doi` equal to the deposit DOI.

Excluded. AIST SDBS (view-only; no bulk export) [National Institute of Advanced Industrial Science and Technology, 2026]; NIST WebBook join code exists in the repository but contributes **0** records to the released IReXP (`ir_source` is always `experimental` for released rows).

2.2 Discovery and fetch (released corpus)

1. Discover IR-reporting OA PMCIDs via NCBI `esearch` (`open access[filter] + characterisation-format IR query`; year/month sliced).
2. Fetch OA full text from PMC-OA S3 plain-text objects (`PMC{id}.{v}.txt`).
3. Ingest Chemotion deposit author-curated band lists + structures (`scripts/chemotion_to_irexp.py`).
4. Persist harvest provenance in `seen_papers.txt.gz` and optional rawer rows in `ir_harvest_snapshot.jsonl.gz` (134,893 rows including intermediate prose fields; the curated release is `irexp.jsonl.gz`).

Non-OA / Scraping fence. Development adapters for ChemRxiv, Beilstein, and generic publisher pages exist for exploration. They are **outside the construction path of the released dataset**. No released `source_doi` is a paywalled publisher DOI. Methods for IReXP as published here cite only **PMC-OA S3 + Chemotion**.

2.3 Extraction (band lists, not spectra)

A deterministic regular-expression pipeline (`spectro_scraper/extract.py`) segments experimental text per compound and extracts:

- IR wavenumbers $\rightarrow$ `ir_bands_cm-1` (list of floats, cm^{-1});
- ^1H and ^{13}C payloads $\rightarrow$ `h_nmr` / `c_nmr` (author strings, when present).

Gates reject scan-range artefacts and common prose false positives. **No curve digitisation** is performed: figures are not traced; JCAMP-DX is not stored.

2.4 Structure resolution

Where an IUPAC or systematic name is available, names are converted with OPSIN [Lowe et al., 2011], canonicalised with RDKit [Landrum, 2026], and encoded as InChIKey and SELFIES [Krenn et al., 2020]. An optional PubChem [Kim et al., 2023] name fallback (`USE_PUBCHEM`) handles trivial names. Structure coverage of the full release is **35.5%** (43,060 / 121,233). The structure-complete split is shipped as `irexp_resolved`.

2.5 Licence handling

Pool	Records	Stamp
commercial (CC-BY + CC0)	87,617	<code>license_pool=commercial;</code> Zenodo / Sci Data primary
non_commercial (CC-BY-NC*)	20,938	held aside
sharealike (Chemotion + rare PMC SA)	1,897	CC-BY-SA / CC-BY-SA-4.0
empty_unknown	10,781	excluded from commercial deposit

`scripts/join_pmc_licences.py` stamps every row; `scripts/split_license_pools.py` reports provenance and materialises pool files under `data/irexp/licence_pools/`. Narrative and policy: `docs/scientific_data/LICENCE_REMEDIATION.md`.

2.6 Quality tooling

Optional physics gates live in `spectro_scraper/quality.py` (^{13}C peak count $\leq$ carbon count; ^1H integration vs formula; IR wavenumber windows). They were **not** applied as a hard filter at harvest; Technical Validation reports a full-corpus post-hoc quarantine on `irexp_resolved` (`scripts/quarantine_structure_nmr.py` $\rightarrow$ `data/audit/structure_nmr_quarantine.jsonl.gz`). Transcription and recall-proxy scripts: `scripts/audit_extraction.py`, `scripts/audit_extraction_recall.py`.

3 Data Records

3.1 Object definition

Each IRexp record is a JSON object. Required chemistry fields for an IR entry:

Field	Type	Description
<code>id</code>	string	Stable record id (InChIKey when resolved, else internal)
<code>ir_bands_cm-1</code>	float list	Experimental IR peak positions (cm^{-1})
<code>ir_source</code>	string	experimental for all released rows
<code>source_doi</code>	string	PMC:<pmcid> or Chemotion deposit DOI
<code>h_nmr / c_nmr</code>	string or null	Author-reported shift lists
<code>smiles / selfies / inchikey</code>	string or null	Resolved structure encodings
<code>has_structure</code>	bool	Convenience flag
<code>license / license_pool</code>	string	Per-article stamp (Europe PMC join or Chemotion)

Not included: absorbance traces, intensities, instrument metadata beyond what appears in source text, PDFs, figures, or full article bodies.

3.2 Files and counts

Paths relative to the project repository / Hugging Face mirror.

File	Records	Description
<code>data/irexp/irexp.jsonl.gz</code>	121,233	Full curated release
<code>data/irexp_resolved/irexp_resolved.jsonl.gz</code>	43,060	100% structure-linked
<code>data/irexp_release/train_no_bench.jsonl.gz</code>	42,808	Resolved minus IRSpectra-Bench InChIKeys
<code>data/irexp_release/train_no_bench_nmr.jsonl.gz</code>	32,949	Same with both ^1H and ^{13}C
<code>data/irexp_release/pretrain_ir.jsonl.gz</code>	119,345	PMC-only IR pretrain pool
<code>data/irexp/seen_papers.txt.gz</code>	188,016 lines	PMC IDs scanned at harvest
<code>data/irexp/ir_harvest_snapshot.jsonl.gz</code>	134,893	Intermediate harvest snapshot
<code>licence_pools/irexp_commercial.jsonl.gz</code>	87,617	CC-BY + CC0 (Zenodo/Sci Data primary)
<code>licence_pools/irexp_non_commercial.jsonl.gz</code>	20,938	NC* held aside
<code>licence_pools/irexp_sharealike.jsonl.gz</code>	1,897	Chemotion + rare PMC SA
<code>licence_pools/irexp_empty_unknown.jsonl.gz</code>	10,781	Excluded from commercial

Composition of irexp.jsonl.gz.

Slice	Count
All IR band-list records	121,233
With ^1H and/or ^{13}C NMR	87,075 (72%)
With resolved structure	43,060 (35.5%)
Structure + any NMR	40,702
Full IR + ^1H + ^{13}C + structure	33,201
PMC OA provenance	119,345
Chemotion provenance	1,888
Unique PMC accessions	15,416

Licence-pool counts (confirmed Track 1 join, 2026-08-26). Lookup over **15,416** unique PMCIDs; stamped pool files under `data/irexp/licence_pools/`. Sum of pools = **121,233** (unchanged).

Pool file	Count	Notes
<code>irexp_commercial.jsonl.gz</code>	87,617	CC-BY / CC0 — Zenodo primary
<code>irexp_non_commercial.jsonl.gz</code>	20,938	CC-BY-NC* held aside
<code>irexp_empty_unknown.jsonl.gz</code>	10,781	empty / unresolved — excluded from commercial deposit
<code>irexp_other.jsonl.gz</code>	0	reserved (e.g. CC-BY-ND)
<code>irexp_sharealike.jsonl.gz</code>	1,897	Chemotion CC-BY-SA-4.0 (1,888) + rare PMC SA

The full `irexp.jsonl.gz` remains multi-licence on disk; commercial redistribution must use the commercial pool (or `license_pool == "commercial"`). Hugging Face was re-issued with stamped pools and honest card text (`scripts/publish_hf.py`, 2026-08-26).

Median bands: **9** (PMC), **39** (Chemotion). All **1,360,866** released IR band values fall inside $350\text{--}4000\text{ cm}^{-1}$ (full-corpus range check; Technical Validation).

3.3 Access

- **Hugging Face:** <https://huggingface.co/datasets/ilkhamfy/IRexp>
- **GitHub development mirror:** <https://github.com/IlkhamFY/spectro-agent>
- **Zenodo archival snapshot:** DOI pending mint after honest multi-licence metadata.

4 Technical Validation

Machine-readable package: `docs/scientific_data/qc_structure_nmr.json` (and artefacts under `data/audit/`).

4.1 Transcription fidelity (PMC)

On a seed-fixed random sample of **60** PMC-sourced records (`scripts/audit_extraction.py --n 60 --seed 0`), each article was re-fetched from Europe PMC and every recorded wavenumber was checked against the source text ($\pm 1\text{ cm}^{-1}$ integer match). Result: **560/560 bands** and **60/60 records** confirmed. Enlarged sample **n=200** (same script/seed family, `data/audit/extraction_audit_n200.json`): **2,250/2,261 bands (99.51%)** and **196/200 records (98.0%)** fully confirmed. This bounds **transcription** error (hallucinated, mis-parsed, or unit-mangled values).

4.2 Extraction-recall proxy (PMC, harvest path)

A human mark-up of every IR string in every paper remains the gold standard. As an automatic **proxy** on the actual harvest path (`scripts/audit_extraction_recall.py --n 40 --seed 0`): re-fetch PMC-OA S3 plain text for **40** distinct source PMCID; re-run `extract_records`; compare band-sets to released rows ($\pm 1\text{ cm}^{-1}$). Result: **2,640/2,640** released bands confirmed in re-extract; **248/248** released IR lists recovered (list-level recall proxy **1.0**; 40/40 papers); **4/40** papers yielded *extra* re-extracted lists relative to the curated release (parser over-fire vs post-harvest dedup/curation, not missed released compounds). Summary: `data/audit/extraction_recall_proxy.json`. This is **not** a substitute for expert human recall.

4.3 Structure-NMR consistency and quarantine (resolved)

Sample (prior). On **500** `irexp_resolved` records with ^{13}C text (seed 0): **17/500 (3.4%)** listed more peaks than carbons. On **500** with ^1H text: integrals > formula H+2 in **17/497 (3.4%)**.

Full resolved corpus. `scripts/quarantine_structure_nmr.py` applied the same physics gates to all **43,060** structure-linked rows. **1,882 (4.37%)** fail ≥ 1 hard check and are listed in `data/audit/structure_nmr_quarantine.jsonl.gz` (diagnostic only — release files unchanged). Among rows with the relevant modality: ^{13}C peaks > carbons **1,194/34,231 (3.49%)**; ^1H integral > formula+2 **1,141/39,672 (2.88%)**; IR out-of-range **0**; unparseable SMILES **0**. Re-users should filter quarantined IDs before supervised training. These rates are integrity diagnostics, **not** an expert skeleton audit.

4.4 IR physical window

Every band in the full 121,233-record release lies in **[350, 4000] cm^{-1}** (0 / 1,360,866 out of range).

4.5 QC status

Check	Status
Per-PMCID licence join + pool counts	Done — commercial 87,617
Transcription fidelity n=60 and n=200	Done
Extraction-recall automatic proxy (n=40 papers, S3 path)	Done (human recall still optional)
Full-corpus structure-NMR quarantine	Done — 1,882 / 43,060 flagged
Expert structure spot-check ($n \geq 100$)	Optional / deferred

5 Usage Notes

- **Band lists $\neq$ spectra.** Do not evaluate models trained on IRexp as if they had seen full absorbance curves.
- **Separate pools by density and licence.** PMC (sparse, author-transcribed) vs Chemotion (denser, author-curated ELN band lists). Keep Chemotion ShareAlike constraints in mind when combining pools; combined redistribution of Chemotion-derived rows must honour CC-BY-SA-4.0.
- **Do not assume PMC = CC-BY-4.0.** Filter to `license_pool == "commercial"` (or use `irexp_commercial.jsonl.gz`) for commercial redistribution; attribute via `source_doi / pmcid`.

- **Structure–NMR quarantine.** Before supervised training on `irexp_resolved`, optionally drop IDs in `data/audit/structure_nmr_quarantine.jsonl.gz` (~4.4% of resolved rows).
- **Training without benchmark leakage.** If using the complementary `IRSpectra-Bench` problems, fine-tune from `train_no_bench.jsonl.gz` (or rebuild with `scripts/build_train_no_bench.py`) so benchmark InChIKeys are withheld.
- **Structure coverage.** Prefer `irexp_resolved` for supervised structure tasks; 64.5% of records lack SMILES.
- **Attribution.** Cite this Data Descriptor / Zenodo DOI and attribute originating articles through each record’s `source_doi`.

6 Data Availability

IRexp numeric extracts are available at:

- Hugging Face Datasets: <https://huggingface.co/datasets/ilkhamfy/IRexp>
- GitHub: <https://github.com/IlkhamFY/spectro-agent> (data/irexp/, data/irexp_resolved/, data/irexp_release/)
- Zenodo: archival DOI at proof; primary artifact = commercial pool, cc-by-4.0 metadata + SA companion
Licensing summary (honest):
- **Chemotion (1,888):** CC-BY-SA-4.0 [Jung et al., 2024].
- **PMC (119,345):** mixed Creative Commons — stamped per article; commercial redistributable **87,617** (CC-BY/CC0); NC* **20,938** held aside; empty/unknown **10,781** excluded from commercial Zenodo (`LICENCE_REMEDIATION.md`).
- Only extracted numeric fields and identifiers are redistributed; source full texts are not.

7 Code Availability

Harvesting, extraction, structure resolution, licence pool splitting, and validation scripts are in <https://github.com/IlkhamFY/spectro-agent> under the **MIT License** (`LICENSE`). Principal modules: `spectro_scraper/` (`extract.py`, `normalize.py`, `pipeline.py`, `quality.py`); release harvest `scripts/s3_ir_harvest.py`, `scripts/chemotion_to_irexp.py`; validation `scripts/audit_extraction.py`, `scripts/audit_extraction_recall.py`, `scripts/quarantine_structure_nmr.py`; pool split `scripts/split_license_pools.py`. The version corresponding to this Descriptor will be tagged at Zenodo deposit time.

8 Author contributions

I.Y.: conceptualization, methodology, software, data curation, validation, writing — original draft.

R.S.: methodology, software, validation, writing — review and editing.

R.A.V.-H.: conceptualization, supervision, writing — review and editing.

9 Competing interests

The authors declare no competing interests.

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References

- Nicole Jung, Pierre Tremouilhac, Dev Punjabi, and Pei-Chi Huang. Chemotion Repository – data collection: FT-IR spectroscopy data (Chemotion IR). Technical report, Karlsruhe Institute of Technology, 2024.
- Sunghwan Kim, Jie Chen, Tiejun Cheng, Asta Gindulyte, Jia He, Siqian He, Qingliang Li, Benjamin A. Shoemaker, Paul A. Thiessen, Bo Yu, Leonid Zaslavsky, Jian Zhang, and Evan E. Bolton. PubChem 2023 update. *Nucleic Acids Res.*, 51:D1373–D1380, 2023. doi: 10.1093/nar/gkac956.
- Mario Krenn, Florian Häse, AkshatKumar Nigam, Pascal Friederich, and Alan Aspuru-Guzik. Self-referencing embedded strings (SELFIES): a 100% robust molecular string representation. *Mach. Learn.: Sci. Technol.*, 1:045024, 2020. doi: 10.1088/2632-2153/aba947.
- Greg Landrum. RDKit: Open-source cheminformatics. <https://www.rdkit.org>, 2026. Accessed 2026-01-01.
- Daniel M. Lowe, Peter T. Corbett, Peter Murray-Rust, and Robert C. Glen. Chemical name to structure: OPSIN, an open source solution. *J. Chem. Inf. Model.*, 51:739–753, 2011. doi: 10.1021/ci100384d.
- National Institute of Advanced Industrial Science and Technology. Spectral Database for Organic Compounds (SDBS). <https://sdb.sdb.aist.go.jp>, 2026. Accessed 2026-01-01.
- National Institute of Standards and Technology. NIST Chemistry WebBook, SRD 69. <https://webbook.nist.gov>, 2026. doi:10.18434/T4D303; accessed 2026-01-01.
- National Library of Medicine. NCBI PMC Open Access Subset. <https://www.ncbi.nlm.nih.gov/pmc/tools/openftlist/>, 2026. Accessed 2026-01-01.
- Jun-Jie Wang, Yongqi Jin, Chen-Yu Zhi, Yu-Jie Liu, Xu-Hao Huang, Fanjie Xu, Xiaohong Ji, Xi Fang, Haoyi Tao, Weinan E, Linfeng Zhang, Guolin Ke, and Rong Zhu. NMRExp: a database of 3.3 million experimental NMR spectra. *Sci. Data*, 12:1956, 2025. doi: 10.1038/s41597-025-06245-5.
- Ilkham Yabbarov, Rudra Sondhi, and Rodrigo A. Vargas-Hernández. Generation recall, not verification, binds LLM structure elucidation from literature spectra. *ICLR*, 2026. Conference manuscript, in preparation.
- Liujia Yang, Zhuo Yang, Jiaqing Xie, Yubin Wang, Ben Gao, Xingjian Wei, Jiaying Sun, Jiang Wu, Conghui He, Yuqiang Li, and Qinying Gu. NMRTrans: structure elucidation from experimental NMR spectra via set transformers. In *Proc. 32nd ACM SIGKDD Conf. Knowledge Discovery and Data Mining (KDD '26)*, Vol. 2, pages 12621–12632, 2026. doi: 10.1145/3770855.3818935. arXiv:2602.10158; NMRSpec corpus context.
- Federico Zipoli, Marvin Alberts, and Teodoro Laino. IR-NMR multimodal computational spectra dataset for 177K patent-extracted organic molecules. *Sci. Data*, 12:1380, 2025. doi: 10.1038/s41597-025-05729-8.